

Hardware Interface Summary

PS configuration, PL interfaces and host interface

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Scope

This document summarises the hardware configuration the firmware and the host software depend on. It was compiled from the files in this folder (01_PS_Configuration/psu_init.c, psu_init.html, 02_Vitis_Platform/export.json) and from the hardware identifiers the firmware uses (FF/01_Source/platform_ids.h, pcie_regs.h). The authoritative PL description is the hardware hand-off H2C_pcie_control.xsa (design_1.hwh). It is listed in 03_PL_Design, to be supplied with the board design.

Device

Item	Value
Device	AMD/Xilinx Zynq UltraScale+ RFSoc xczu47dr-ffvg1517-2-i
Processing system	Quad Cortex-A53 (firmware on A53-0, EL3, standalone), dual Cortex-R5F, PMU
RF data converter	Gen 3 RF-ADC and RF-DAC tiles, 14-bit
Vitis platform	zu47dr_plat, system zu47dr_app_system, application zu47dr_app, domain standalone_domain (export.json, Vitis 2022.2). The firmware also builds under Vitis 2025.2 (SDT).

PS clocking (psu_init)

Clock	Frequency
PSS_REF_CLK	33.333 MHz
APLL / DPLL / RPLL	2400 MHz
VPLL / IOPLL	3000 MHz
ACPU (A53)	1200 MHz
DDR	DDR4-2400
PL0 (FCLK0)	100 MHz
PL1 (FCLK1)	300 MHz
PCIe reference (GT lane 0, refclk0)	100 MHz; PCIe core clock 250 MHz
DisplayPort (GT lane 2)	27 MHz reference

DDR memory

Item	Value
Type	DDR4-2400R, 64-bit bus with ECC
Components	16 Gb, ×16
High address limit	0x1_FFFF_FFFF

MIO assignment (psu_init)

MIO	Function
0 – 5	QSPI (boot flash)
6 / 7	UART0 : firmware console, 115200 8N1
10 / 11	I2C0
13 – 23	SD0
24 / 25	I2C1
26 – 37	GEM0 (RGMII)
40	PCIe reset_n
52 – 63	USB0
64 – 75	GEM3
76 / 77	MDIO

PL interfaces used by the firmware

The names below are the XPAR_<name>_BASEADDR / _DEVICE_ID stems that platform_ids.h resolves; they come from the Vivado block-

design instance names. Under the SDT flow the BSP provides `XPAR_<instance>_BASEADDR`; under the classic flow it provides `_DEVICE_ID`.

Instance	Driver	Function
XRFDC_0 / USP_RF_DATA_CONVERTER_0	XRFdc	RF-ADC / RF-DAC tiles: rates, NCO, decimation, interpolation, DSA, VOP, QMC, MTS
ADC_CHANNEL_SWITCH_ADC_SWITCH	XAxisScr	ADC AXIS routing, 8 inputs → 4 outputs. Default 0x1237 (ch0=7, ch1=3, ch2=2, ch3=1).
DDS_H2C_SWITCH	XAxisScr	DAC input select. S00 = host H2C (External_Interface_H2C_data_in), S01 = DDS (dds_compiler_2) ; M00 → axis_broadcaster_1 → DAC tiles
ADC_CHANNEL_SELECT	XGpio	PL GPIO channel mode: 1 = one, 2 = two, 3 = four channels into C2H
AXI_GPIO_DYNAMIC_CLK	XGpio	Dynamic stream clock control
CLOCK_COUNTER	XGpio	Stream / PL clock measurement (menu 11)
AXI_GPIO_DDS1DDS2	XGpio	DDS control (two GPIO channels, released at init)
SYSREF_CONTR0LL	XGpio	SYSREF control for MTS
QUAD_SPI_LMK	XSpi	LMK04828 clock distributor
QUAD_SPI_ADC_LMX, QUAD_SPI_DAC_LMX	XSpi	LMX2594 synthesizers, ADC and DAC
CLK_WIZ_ADC, CLK_WIZ_DAC	raw AXI-Lite	Stream clocks. The DAC stream is 200 MHz: one 32-bit I/Q word per clock = 762.9 MiB/s.
M00_PCIE_CONTROL (base 0xA00A0000, 64 KB)	raw	PCIe control mailbox. Host side: BAR 2. ring_register at 0xA8, RFDC_configuration at 0xAC.

Host interface (PCIe / QDMA)

Item	Value
DMA engine	AMD/Xilinx QDMA in the PL, Linux driver <code>iwfg</code>
Device nodes	<code>/dev/iwfg0</code> = C2H (capture), <code>/dev/iwfg1</code> = H2C (transmit)
Control	BAR 2 register mailbox (ABI 4)
C2H stream	1, 2 or 4 channels × 200 MSPS × 4 B (Cs16) = 763 / 1526 / 3052 MiB/s
H2C stream	762.9 MiB/s continuous. The DAC input has no TREADY.

PL hand-off files (to be supplied)

These files belong to the board design and are not in this package. Place them in `03_PL_Design` when you archive a build:

File	Used by
H2C_pcie_control.xsa	Vitis platform creation (BSP, <code>xparameters.h</code>)
H2C_pcie_control.bit	BOOT.BIN, PL partition. It must precede the application.
design_1.hwh	Reference for the switch port wiring (HW_DAC_SW_SI_HOST/DDS in <code>platform_ids.h</code>)
fsbl.elf, pmufw.elf	BOOT.BIN, generated by the platform build